

ECE391: Computer Systems Engineering
Machine Problem 3

Spring 2025
Checkpoint 1: Mon April 7 18:00 CDT
Checkpoint 2: Fri April 18 18:00 CDT

MP3 Student Rubric

Checkpoint 1 (0-30pts)

1. Cache (6pts)
 - Cache is empty on OS startup
 - Starting with an empty cache, can induce 64 misses on different blocks and then 64 read hits on those same blocks with a total of 64 requests made to the backing device
 - Must not generate a request to the backing device on a read hit
 - No evictions unless `cache_get_block` is called again
2. KTFS (10pts)
 - Open/Close
 - Readat
 - Ioctl (GETEND)
3. Vioblk (7pts)
 - Open/Close
 - Readat
 - Writeat
 - Ioctl (GETEND, GETBLKSZ)
4. ELF Loader (4pts)
 - Load a valid ELF file into appropriate memory and return the entry point
 - Reject invalid ELF files with an error code (magic bytes, class, endianness, version, type, architecture, etc.)
5. Memio (2pt)
 - Readat
 - Writeat
 - Ioctl (GETEND, GETBLKSZ, SETEND)
6. Locks (1pt)

Penalties

1. Unable to check out `main` branch (-30pts)
2. Unable to compile `main` branch (-30pts)
3. Compiler warnings when compiling `kernel.elf` (-5pts)

Checkpoint 2 (0-40pts)

1. Virtual Memory (10pts)
 - Able to correctly map pages with the associated flags
 - Able to correctly map ranges with the associated flags
 - Able to correctly unmap ranges with the associated flags
 - Functional lazy allocation
2. System Calls (20pts)
 - `sysexit`
 - `sysprint`
 - `sysdevopen`, will be tested indirectly through the usage of read/write syscalls
 - `sysfsopen`, will be tested indirectly through the usage of read/write syscalls
 - `sysclose`
 - `sysread`
 - `syswrite`
 - `sysioctl`
 - `sysexec`
 - `sysusleep`
 - `sysfscreate`
 - `sysfsdelete`
3. KTFS (10pts)
 - `Writeat`
 - `Create`
 - `Delete`

Penalties

1. Unable to check out `main` branch (-40pts)
2. Unable to compile `main` branch (-40pts)
3. Unable to run the provided `main.c` template (-40pts)
4. Compiler warnings when compiling `kernel.elf` (-5pts)

Checkpoint 3 (0-30pts)

1. `syswait`
 - Tested via an user program
 - 2 points
2. Smooth Forking
 - Can run 16 processes, exit 16 processes, and run 16 processes again via forking
 - 5 points
3. `pipes`
 - `pipes` testcase (u mode??)
 - 3 points
4. Shell support
 - Shell Execution: Can run all of the released cp2 user programs using shell without no crashing
 - 5 points
 - Shell Commands: Full support for all shell commands/inputs
 - 5 points
5. Forking
 - Can run `fib` testcase
 - 5 points
6. 2 Trek
 - The `init` testcase released launches two working treks
 - 5 points

Penalties

1. Unable to check out `main` branch (-30pts)
2. Unable to compile `main` branch (-30pts)
3. Compiler warnings when compiling `kernel.elf` (-5pts)